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CRETACEOUS AND EARLY TERTIARY PALEOMAGNETIC RESULTS FROM THE ANTARCTIC PENINSULA

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Abstract. Paleomagnetic results are presented for early Tertiary and Cretaceous igneous and volcanic units from the northern Antarctic Peninsula. When combined with previously published data, they permit the construction of a polar wander path that is very similar to one derived for the East Antarctic craton by rotating Tertiary and Cretaceous poles from India and Australia into the Antarctic frame of reference. When considered in this context, the paleomagnetic data from the Orville Coast, which was previously interpreted as evidence in favor of oroclinal bending or relative rotation of the southern Antarctic Peninsula with respect to the northern part, may be considered as an in situ result. The implications of the new data are as follows: (1) The Antarctic Peninsula down to the Orville Coast has been a part of the Antarctic plate since 100 m.y. ago. (2) Movement of the Antarctic Peninsula required to avoid overlap in reconstructions of Gondwanaland must have occurred before 100 m.y. ago.

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INTRODUCTION

Detailed paleomagnetic data from the Antarctic Peninsula are potentially useful in resolving several problems in Antarctic tectonics including the place of the Antarctic Peninsula in reconstructions of Gondwanaland, the reality of oroclinal bending in both the northern and southern parts of the peninsula, and the role of this crustal unit in the evolution of the south eastern Pacific. The last topic is particularly relevant in view of the possibility that a plate boundary existed within the confines of the present Antarctic plate during the Cretaceous and early Tertiary (Molnar et al., 1975; Suarez and Molnar, 1980; Gordon and Cox, 1980). Presented here are new data derived from volcanic and intrusive rocks of Cretaceous and Tertiary age collected during the 1978-1979 and 1980-1981 austral field seasons. Combined with previously published data, the paleomagnetic results are of sufficient quality and density to construct a tentative apparent polar wander path for the middle Cretaceous through early Tertiary. An apparent polar wander path for the east Antarctic craton is constructed by using paleomagnetic poles from India and Australia compiled by Klootwijk and Peirce (1979) and sea floor spreading reconstructions of Norton and Sclater (1979) with modifications on the separation of Australia from Antarctica suggested by Cande and Mutter (1982). The above problems

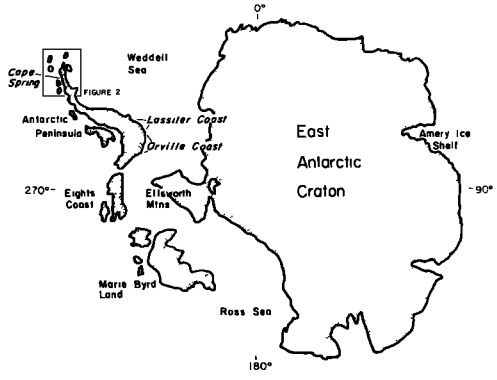


Fig. 1. Generalized map (from Elliot, 1975) showing major crustal units of Antarctica, and pertinent geographic localities.

are addressed by a comparison of these two polar wander paths in the present Antarctic reference frame.

PREVIOUS STUDIES

Blundell (1962) carried out the first paleomagnetic study in the Antarctic Peninsula, measuring the natural remanent magnetization (NRM) of samples collected from intrusive and volcanic units ranging in age from Jurassic (?) to Pleistocene. A mean paleomagnetic pole was calculated for the Andean Intrusive Suite which was interpreted to be representative of the Cretaceous paleomagnetic field with respect to Antarctica.

Hamilton (1966) reinterpreted Blundell's (1962) results as evidence for oroclinal bending of an originally straight Antarctica Peninsula.

Dalziel et al. (1973) published paleomagnetic results from 40 m.y. intrusives collected from Livingston Island, South Shetland Islands (Figure 1 and Figure 2). They suggested this result was consistent with no post-40 m.y. bending and that a critical selection of data from Blundell (1962) left the problem of pre-40 m.y. bending unresolved.

Kellogg and Reynolds (1978) reported paleomagnetic results from dikes and plutons of the Lassiter Coast (Figure 1). Of the 17 intrusives with an interpretable paleomagnetic record, three were directly dated by the K-Ar method with an average age of 99 m.y. These data were compared with those of Blundell (1962) and Dalziel et al. (1973) to demonstrate that no detectable postintrusive oroclinal bending of the northern Antarctic Peninsula had occurred, but a rigid rotation of this unit with respect to the East Antarctic craton could not be precluded.

Valencio et al. (1979), using new data from King George (25 de Mayo) Island of "late Mesozoic/early Tertiary" age and from the Cape Spring intrusives (Figure 1) with an average K-Ar age of 98 m.y., also argued against oroclinal bending of the northern Antarctic Peninsula. They demonstrated that the existing data are inconsistent with a model of bending generated by poles of rotation within the Peninsula.

To investigate the possibility that the southern Antarctic Peninsula has undergone rotation with respect to the northern, Kellogg (1980) studied the paleomagnetic record of intrusives from the Orville Coast and eastern Ellsworth Land (Figure 1). Of the 17 units with interpretable paleomagnetic results, two are dated by the Rb-Sr isochron method with an average age of 106 m.y. Kellogg noted that the mean paleomagnetic declination of the Orville Coast plutons is 51° east of the Lassiter Coast mean declination and interpreted this as evidence for oroclinal bending or relative rotation of the southern Antarctic Peninsula with respect to the Lassiter Coast.

Watts (1982) reported results from early Tertiary plugs and lavas from King George Island and computed a preliminary early Tertiary pole. Comparison of this pole with early Tertiary Australian paleomagnetic poles rotated into the Antarctic reference frame also indicated no detectable motion relative to the East Antarctic craton.

Suarez and Molnar (1980) and Gordon and Cox (1980) speculated on the existence of a plate boundary between East and West Antarctica during the Cretaceous and early Tertiary. This is one possible explanation for the observed discrepancy between Pacific plate poles and contemporaneous poles either from the global data set or the Indian plate when rotated into the same frame of reference. As discussed by these workers, an equally plausible explanation is an extinct plate boundary within the confines of the Pacific plate. Gordon and Cox (1980) gave a possible model of relative motion between East and West Antarctica that involved rotation of the Antarctic Peninsula with respect to the East Antarctic craton during the early Tertiary.

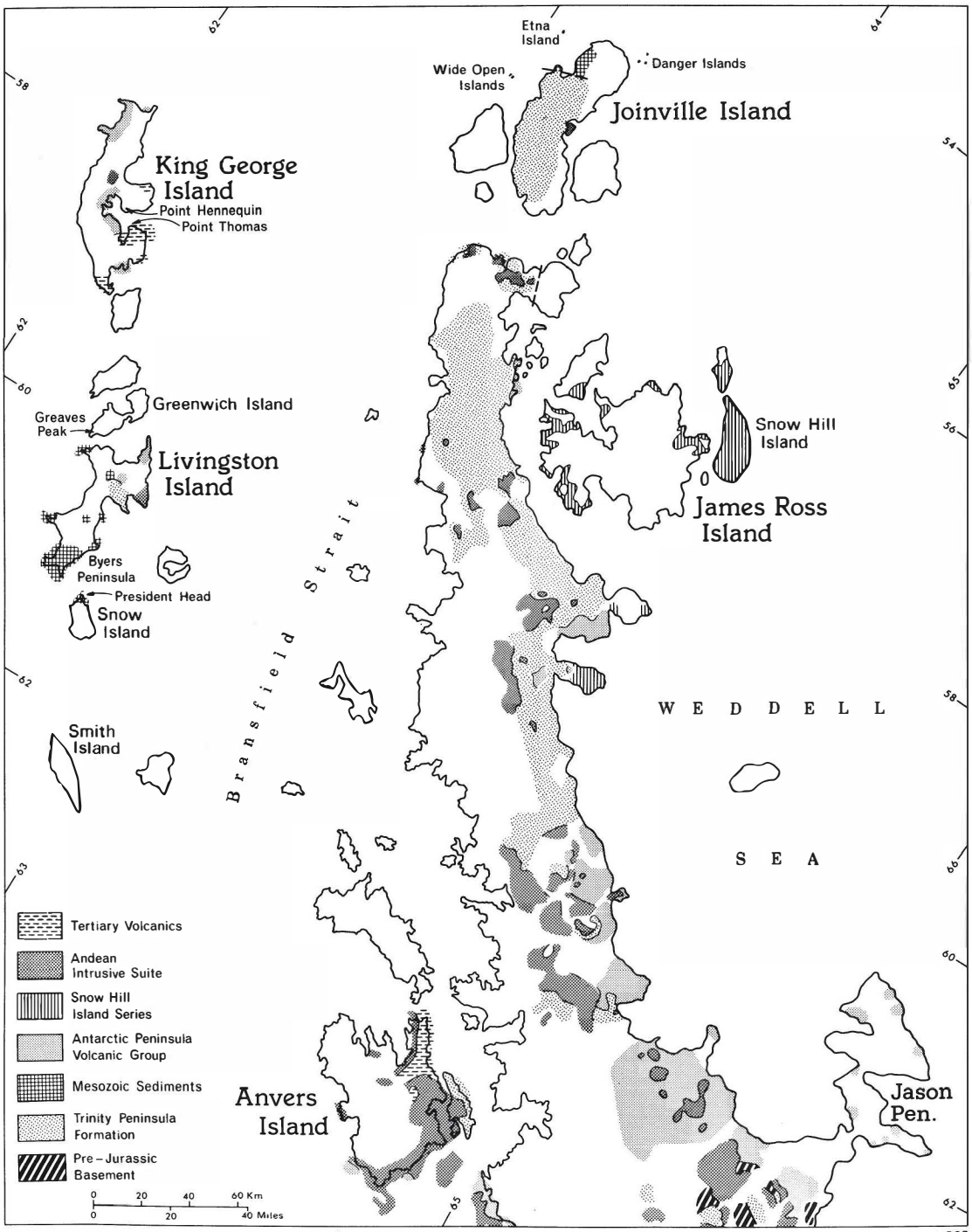


Fig. 2. Generalized geological map of the northern Antarctic Peninsula (after Adie, 1969) showing localities of the paleomagnetic sampling sites. These include the Danger Islands, Etna Island, Wide Open Islands, King George Island, Greenwich Island, Livingston Island, and Snow Island.

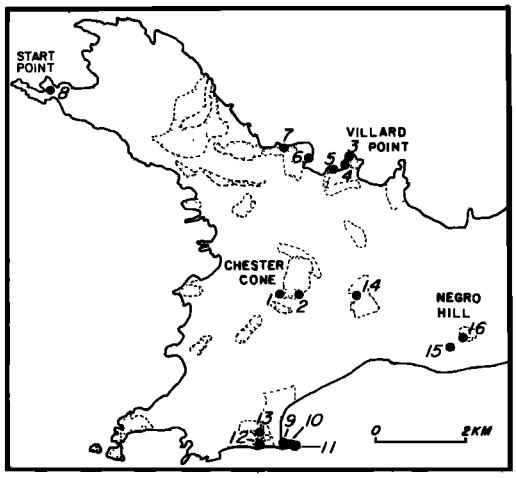


Fig. 3. Generalized geologic map of Byers Peninsula, Livingston Island (after Pankhurst et al., 1980) showing paleomagnetic sampling sites. The stipple shows major intrusives or plugs and the dashed lines indicate positions of lavas.

NEW DATA

The new data presented here are derived from volcanic and intrusive rocks exposed on various offshore islands depicted in Figure 2. From the South Shetland Islands, the early Tertiary Ezcurra Inlet volcanic group, and the Point Hennequin volcanic group were sampled as were early Tertiary sills from President Head, Snow Island. Results probably representative of the Cretaceous paleomagnetic field are derived from a layered basic intrusive from Greaves Peak, Greenwich Island, and from igneous rock units exposed on Byers Peninsula, Livingston Island. Results are also derived from the undated Danger Islands gabbro, Etna Island diorite, and Wide Open Islands gabbro.

Sixteen samples from Point Hennequin and eight samples from the Danger Island gabbro were block samples oriented with a Brunton compass and clinometer. All other samples were oriented drill cores, always oriented with a Brunton compass and clinometer, but magnetic declinations were checked where possible with a direct reading solar compass.

The Ezcurra Inlet group and the Point Hennequin group are successions of agglomerate, andesite and tuff (Barton, 1966). Seventy-three oriented cores were collected from twelve lavas, two dikes, and one volcanic plug (Jardine Peak) along the coast of Ezcurra Inlet, King George Island. Watts (1982) reported paleomagnetic data and K-Ar whole rock ages from three lavas and a volcanic plug assigned to the same group but from a different locality on King George Island. The K-Ar ages ranged from 58 to 49 m.y. with an average of 52 m.y. Dupre (1982) has determined Ar^{40}-Ar^{39} ages ranging from 52 to 42 m.y. with an average of 48 m.y. for various units from the Ezcurra Inlet group.

Sixteen block samples were collected from four sites in the Point Hennequin group, which has a single Ar^{40}-Ar^{39} age of 47 m.y. (Dupre, 1982). The average age of the Point Hennequin and Ezcurra Inlet group volcanics is 49 m.y.

Fifty oriented cores were collected from eight basic sills and a rhyolite lava on President's Head, Snow Island. One of the sills has been dated by Dupre (1982) giving an Ar^{40}-Ar^{39} age of 54 m.y.

Twenty oriented cores were collected from four sites in a layered basic intrusive exposed in the vicinity of Greaves Peak, Greenwich Island. Dupre (1982) reports an Ar^{40}-Ar^{39} age of 82 m.y.

A total of ninety-two cores was collected from sixteen sites in plugs, lavas and a dike from Byers Peninsula, Livingston Island (Figure 3). Sites 1 and 2 are from the Chester Cone plug dated at 132 m.y. by Pankhurst et al. (1980) using the K-Ar method on whole rock and mineral separates. Gracanin (1983) confirms this result, obtaining an Ar^{40}-Ar^{39} age of 130 m.y. from a hornblende separate. Site 8 is an andesite dike with a mean whole rock K-Ar age of 128 m.y. (Pankhurst et al., 1980). Sites 9,10, and 11 are each from distinct flows in a sequence with an average Ar^{40}-Ar^{39} age of 104 m.y., and the plug at Sealer Hill, site 13, has an Ar^{40}-Ar^{39} age of 111 m.y. (Gracanin, 1983). Site 14 is a basic lava with a reported K-Ar age of 80 m.y. though Pankhurst et al., (1980) regard this as rather low. Site 15 is a basic sill with an average K-Ar whole rock age of 75 m.y., and site 16 is the Negro Hill plug dated at 92 m.y. by the same method (Pankhurst et al., 1980). Sites 3, 4, 5, 6, 7, and 12 are not dated.

MAGNETIC REMANENCE MEASUREMENT

The samples were cored and sliced into cylinders 2.54 cm wide and 2.3 cm high.

The measurements of the remanent magnetization, demagnetization experiments

and data reduction were carried out in the paleomagnetic laboratories at the University of Michigan and the University of Leeds. During the course of this investigation, we employed a ScT cryogenic magnetometer, a Schonstedt SSM-1A spinner magnetometer, a Digico complete results magnetometer, and the British prototype CCL cryogenic magnetometer. Thermal demagnetization was carried out by using the coil-feedback nulled oven and Schonstedt TSD-1 mu-metal shielded oven at the University of Michigan and a large coil-feedback system and bench type mu-metal oven at Leeds. Alternating field demagnetization was carried out using the Schonstedt CSD-1 static demagnetizer at Michigan and the tumbler demagnetizer (McElhinny, 1966) at Leeds.

After each demagnetization step, the result was plotted on an orthogonal projection (Zijderveld, 1967) in sample coordinates as an aid to choosing the next demagnetization step. The goal of each demagnetization sequence was to determine the components that make up the NRM and to isolate each using the appropriate selection of steps. Components were identified from linear trajectories on the orthogonal projections. Samples which yielded erratic trajectories were considered to have no paleomagnetic information. Samples were transported from the demagnetization devices to the measuring instruments inside mu-metal boxes. Samples were left inside the field free space of the cryogenic magnetometers for up to 5 min to determine if the remanent magnetization showed viscous behavior. If viscous behavior was suspected, the sample was left inside the field free space until such effects had disappeared or it was ascertained that the effect would not disappear over a period of time convenient to the investigation. This behavior was noted only in the medium- to coarse-grained gabbro from the Danger Islands.

Thermal and alternating field demagnetization was carried out on approximately equal numbers of samples, in many cases with one specimen from a 10-cm core treated with alternating field and the other by thermal demagnetization.

PALEOMAGNETIC PROPERTIES

Figure 4 shows orthogonal projections of results of typical demagnetization experiments. Secondary magnetizations were generally removed rather early during AF or thermal demagnetization, giving either linear (WPH 6B, Snow Island sill) or curved (WJA 1B, Wide Open Islands gabbro) trajectories. Characteristic magnetization was then removed at the higher levels of AF and thermal demagnetization yielding linear trajectories descending toward the origin of the orthogonal projections. In most cases, no secondary magnetization was observed and the demagnetization trajectory was essentially linear, beginning from NRM and ending at the origin.

The blocking field spectra and blocking temperature spectra varied from unit to unit and, indeed, from sample to sample. The Danger Islands gabbro had the lowest maximum blocking field with 95% of the magnetization removed at a peak field of 25 mT (Figure 4, WJA 58A). Specimens from the same cores generally had maximum blocking temperatures of 600° (WJA 58B). Specimens from this unit generally exhibited the most pronounced viscous behavior at the highest temperatures, but this was dealt with as described previously.

The lowest maximum blocking temperatures of the characteristic magnetization were observed in specimens from a Snow Island sill at 450° to 500°C. Other specimens from the same unit had maximum blocking fields of 60 to 80 mT. ^{NRM} intensities ranged from 1 to 10^{-2} A/m. No blocking temperatures higher than 600° were observed. Only in one sill from Snow Island was a secondary magnetization found to have substantial blocking temperatures. A component with a maximum blocking temperature of 550°C and normal polarity was removed by thermal demagnetization to reveal a reversed polarity magnetization with blocking temperatures between 550°C and 560°C. A lava from the Ezcurra Inlet group and a sill from Snow Island exhibited complex, unstable behaviour during demagnetization, and these units are regarded as not recording the paleomagnetic field.

The mean paleomagnetic directions statistics and pole positions are given in Table 1. The pole positions were calculated from the mean of the site mean directions.

APPARENT POLAR WANDER PATHS FOR THE ANTARCTIC PENINSULA AND EAST ANTARCTIC CRATON

Selected paleomagnetic poles for the Antarctic Peninsula are given in Table 2. A mean 49 m.y. pole was calculated from

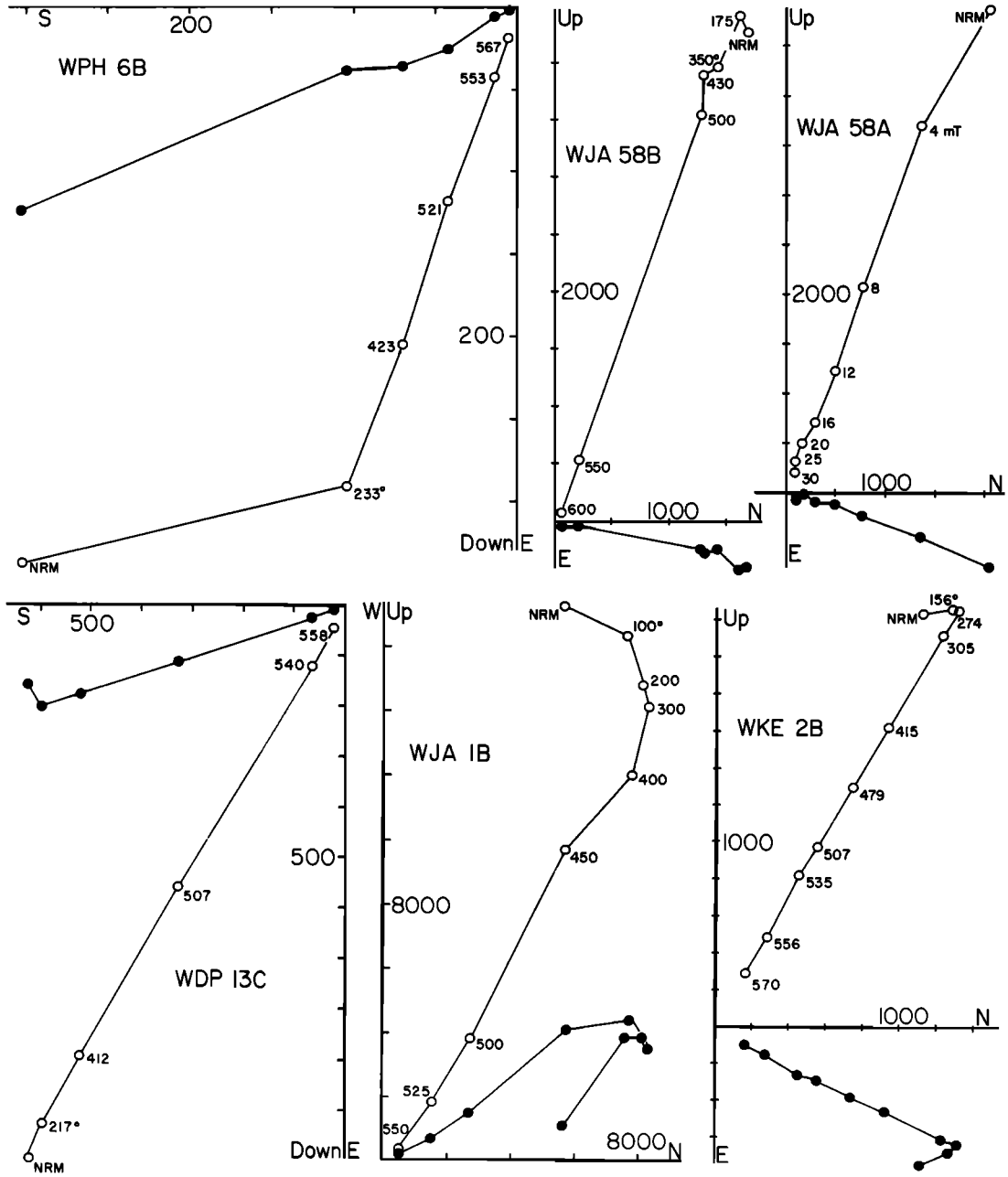


Fig. 4. Orthogonal projections (Zijderveld, 1967) of typical demagnetization results. The open circle denotes projections on to the vertical plane and the closed circle projections on to the horizontal plane. The units on the axes are 10^{-3} A/meter.

the results derived from Ezcurra Inlet group at Ezcurra Inlet, the Ezcurra Inlet group results reported by Watts (1982) and the Point Hennequin group. A second early Tertiary pole was computed from the sills and lavas exposed on President's Head, Snow Island, but the age of magnetization is less well constrained with only a single Ar^{40}-Ar^{39} determination from a sill at 54 m.y.

With only four sites representing the Greenwich Island layered basic intrusive,

TABLE 1. Paleomagnetic Mean Directions, Fishers Statistics and South Pole Directions

Unit/Site Number	N (Af, Therm)	k	α_{95}	Declination	Inclination	Pole Position		dp	dm
						Latitude	Longitude		
Ezcurra Inlet Group									
1 (lava)	8 (5, 3)	894	2°	343°	-66°				
2 (lava)	7 (4, 3)	991	2°	342°	-64°				
3 (lava)	6 (3, 3)	1214	2°	340°	-62°				
4 (lava)	7 (4, 3)	520	3°	344°	-58°				
5 (lava)	5 (2, 3)	1018	2°	304°	-64°				
6 (dike)	7 (4, 3)	271	4°	163°	76°				
7 (lava)	...	...	...	...	...				
8 (lava)	8 (5, 3)	909	2°	326°	-63°				
9 (lava)	5 (3, 2)	33	13°	147°	70°				
10 (dike)	9 (6, 3)	1298	1°	203°	67°				
11 (lava)	8 (5, 3)	137	5°	346°	-51°				
12 (plug)	6 (3, 3)	443	3°	37°	-82°				
13 (lava)	5 (2, 3)	96	8°	160°	58°				
14 (lava)	8 (5, 3)	186	4°	313°	-67°				
15 (lava)	2 (1, 1)	38	42°	355°	-70°				
16+(lava)	11 (11, 0)	55	6°	11°	-69°				
17+(lava)	5 (5, 0)	168	6°	103°	-83°				
18+(lava)	7 (7, 0)	253	4°	54°	-82°				
19+(plug)	6 (6, 0)	52	9°	302°	-72°				
Point Hennequin Group									
1 (lava)	5 (2, 3)	870	3°	187°	71°				
2 (lava)	4 (1, 3)	130	8°	163°	61°				
3 (lava)	4 (2, 2)	577	4°	150°	70°				
4 (lava)	6 (3, 3)	72	8°	153°	62°				
Mean Direction and Pole Ezcurra Inlet, Point Hennequin Group (49 m.y.)	22	42	5°	342°	-70°	79°S	48°E	7.0°	8.2°
President's Head, Snow Island									
1 (sill)	7 (3, 4)	54	8°	148°	70°				
2 (sill)	4 (2, 2)	83	10°	137°	72°				
3 (sill)	7 (3, 4)	474	3°	303°	-78°				
3a (secondary)	4 (0, 4)	201	6°	171°	68°				
4 (sill)	8 (2, 6)	229	4°	331°	-80°				

TABLE 1. (continued)

Unit/Site Number	N (Af, Therm)	k	α_{95}	Declination	Inclination	Pole Position Latitude	Pole Position Longitude	dp	dm
5 (sill)	3 (1, 2)	77	14°	296°	-68°				
6 (sill)	6 (3, 3)	321	4°	288°	-78°				
7 (flow)	7 (2, 5)	144	5°	356°	-75°				
8 (sill)	6 (3, 3)	335	4°	315°	-66°				
9 (sill)	8 (5, 3)	108	5°	326°	-69°				
Mean Direction and Pole Snow Island sills (54 m.y.)	9	117	5°	318°	-74°	61°S	63°E	5.3°	7.2°
Layered Intrusive, Greenwich Island									
1	6 (3, 3)	258	4°	169°	62°				
2	5 (1, 4)	200	5°	161°	60°				
3	7 (2, 5)	89	6°	158°	56°				
4	5 (5, 0)	196	5°	161°	54°				
Mean Direction and Pole Layered Intrusive Greenwich Island (82 m.y.)	4	343	5°	162°	58°	66°S	81°E	5.3°	7.2°
Byers Peninsula									
9 (flow, 104 m.y.)	6 (3, 3)	369	3°	353°	-68°				
10 (flow, 104 m.y.)	8 (5, 3)	415	3°	17°	-63°				
11 (flow, 104 m.y.)	8 (5, 3)	1014	2°	6°	-69°				
13 (plug, 111 m.y.)	7 (4, 3)	189	4°	348°	-71°				
14 (lava, 80 m.y.)	10 (5, 5)	182	4°	47°	-64°				
15 (sill, 75 m.y.)	9 (5, 4)	410	3°	2°	-71°				
16 (plug, 92 m.y.)	6 (3, 3)	271	4°	342°	-55°				
Mean Direction and Pole 96 m.y. igneous units	7	57	8°	5°	-67°	77°S	132°E	11.2°	13.4°
1 (plug, 130 m.y.)	10 (7, 3)	59	6°	356°	-72°				
2 (plug, 130 m.y.)	5 (2, 3)	578	3°	3°	-62°				
3 (intrusion)	12 (6, 6)	723	2°	43°	-68°				
4 (intrusion)	8 (5, 3)	424	3°	292°	-80°				
5 (lava)	7 (4, 3)	54	8°	245°	-82°				
6 (intrusion)	7 (4, 3)	1440	2°	253°	-80°				
7 (intrusion)	9 (4, 5)	197	4°	103°	83°				
8 (dike, 128 m.y.)	13 (8, 5)	84	4°	287°	-65°				

12 (intrusion)	12 (7, 5)	3	31°	33°	-72°				
Wide Open Islands gabbro									
1	4 (2, 2)	53	13°	343°	-56°				
2	6 (3, 3)	113	6°	346°	-56°				
3	5 (1, 4)	66	9°	337°	-56°				
4	6 (2, 4)	131	6°	344°	-58°				
Mean Direction and Pole Wide Open Islands gabbro	4	1169	3°	342°	-56°	62°S	94°E	2.7°	3.8°
Etna Island diorite									
1	11 (7, 4)	90	5°	18°	-73°				
2	4 (3, 1)	562	4°	20°	-71°				
3	3 (2, 1)	95	13°	353°	-68°				
Mean Direction and Pole Etna Island diorite	3	204	9°	9°	-71°	81°S	162°E	13.1°	15.0°
Danger Islands gabbro intrusive									
1	2 (1, 1)	174	19°	12°	-74°				
2	9 (3, 6)	101	5°	356°	-67°				
3	5 (1, 4)	36	13°	333°	-65°				
4	5 (1, 4)	56	10°	12°	-65°				
Mean Direction and Pole Danger Island gabbro	4	97	9°	357°	-68°	78°S	116°E	13.3°	15.7°

N denotes the number of specimens in a site; the numbers in parenthesis are the total treated by alternating field and thermal demagnetization respectively; k is the estimate of Fisher's (1953) precision parameter; α_{95} is the apical half angle of 95% confidence; Declination is the mean declination; Inclination is the mean inclination; latitude is the latitude of the associated pole position; dp and dm are the half angles of the oval of 95% confidence (Creer et al., 1957).

+ Results from Watts (1982).

TABLE 2. Paleomagnetic South Pole Positions, Antarctic Peninsula Pole Positions

Unit	Age m.y.	Latitude	Longitude	α_{95}	dp	dm	Reference
Livingstone Island intrusives	40	77°S	48°E		13.0	15.0	Dalziel et al. (1973)
Ezcurra Inlet-Point Hennequin volcanics	49	79°S	48°E		7.0	8.2	this study
Snow Island sills	54	69°S	13°E		7.6	8.4	this study
Greenwich Island layered intrusive	82	66°S	81°E		5.3	7.2	this study
Cape Spring intrusives	98	86°S	118°E		8.0	9.0	Valencio et al. (1979)
Lassiter Coast intrusives	99	87°S	229°E	11.5			Kellogg and Reynolds (1978)
Orville Coast intrusives	106	71°S	195°E		10.2	11.0	Kellogg (1980)
Byers Peninsula igneous rocks	92	77°S	132°E		11.2	13.4	this study

dated at 82 m.y., there is the possibility that secular variation is not averaged. On the other hand, perhaps the slow cooling of this medium to coarsely crystalline unit provided an averaging mechanism to give the rather high between-site precision.

A mean pole is calculated from the seven sites which range in age from 111 m.y. to 75 m.y. with an average age of 96 m.y. Other sites are either undated or too old to include in this population. The lavas at sites 9, 10, and 11 are dipping by 20° but a structural correction disperses the mean directions implying either remagnetization or a primary dip of the lavas. The dike at site 8 and the undated sites 4, 5, 6, and 7 give poles that fall in rather low latitudes and these rocks may have been magnetized during polarity transitions.

As the Danger Island gabbro, the Wide Open Island gabbro and the Etna Island diorite remain undated, the poles calculated from these units are not used in the construction of the apparent polar wander path.

Of the published paleomagnetic results, those of Blundell (1962) are not used because of the very limited demagnetization carried out. The 40 m.y. pole published by Dalziel et al. (1973) is in excellent agreement with the pole determined from the Ezcurra Inlet-Point Hennequin groups. The pole from the igneous rocks on King George (25 de Mayo) Island reported by Valencio et al. (1979) is not used because of the spread of K-Ar whole rock ages on these rocks from 27 m.y. to 110 m.y. The 98 m.y. pole from the Cape Spring intrusives (Valencio et al., 1979) is indistinguishable from the 99 m.y. pole given by Kellogg and Reynolds (1978) for the Lassiter Coast intrusives.

Although Kellogg (1980) interprets the paleomagnetic results from the Orville Coast as strong evidence supporting oroclinal bending or relative rotation of the southern Antarctic Peninsula with respect to the northern Antarctic Peninsula, for reasons discussed below, the Orville Coast pole is considered as an in situ result.

The dated Antarctic Peninsula poles are listed in Table 2 and plotted in Figure 5.

No Cretaceous or early Tertiary paleomagnetic data are available for the East Antarctic craton, and this polar wander track must be inferred by the rotation of Australian and Indian poles into the Antarctic frame of reference. The poles used in this exercise are those compiled by Klootwijk and Peirce (1979). The poles

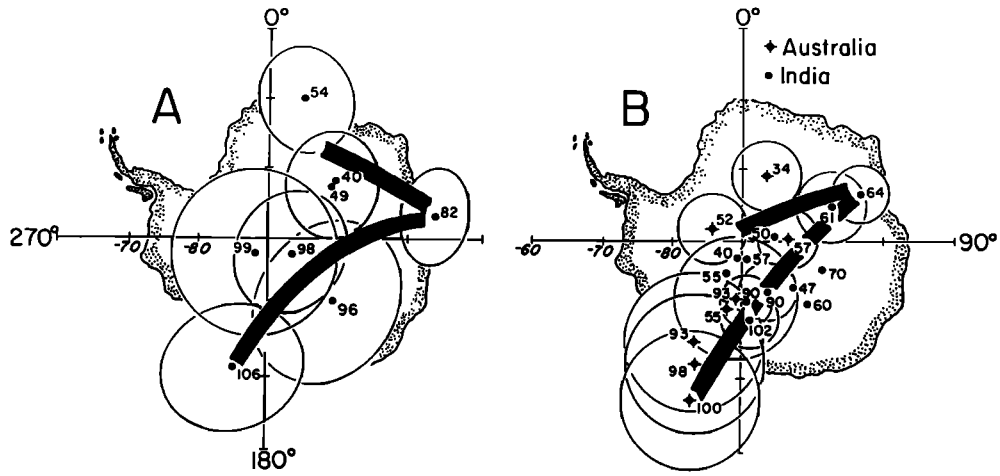


Fig. 5. The apparent polar wander paths for the (a) Antarctic Peninsula and (b) the East Antarctica craton as inferred by rotating poles from Australia (plus) and India (solid circle) into the Antarctic frame of reference. Ovals and circles of 95% confidence are shown except for poles with such figures larger than 12°. The poles are labeled with the inferred age of magnetization.

from the Indian subcontinent were rotated using the parameters given by Norton and Sclater (1979) with interpolations for the ages of the paleomagnetic poles. The parameters of Norton and Sclater (1979) were also used for Australian poles no older than 43 m.y. Following the re-identification of marine magnetic anomalies between Australia and Antarctica by Cande and Mutter (1982) only poles 80 m.y. or older were rotated to the fit of Antarctica to Australia. The position of poles between these age limits was estimated by interpolation according to the age of the pole.

The results of this exercise are summarized in Table 3 and plotted in Figure 5. The path begins with mid-Cretaceous poles in the vicinity of the Ross Sea. The pole apparently migrated across the East Antarctic craton to a position near the Amery Ice Shelf during latest Cretaceous time and then returned to the interior of the craton during Tertiary time.

If the Orville Coast result is considered in situ, it agrees very well with the mid-Cretaceous poles from Australia and India rotated into the Antarctic reference frame in the vicinity of the Ross Sea. Any utilization of paleomagnetic data for the inference of tectonic rotations must take into consideration the geometry of the reference polar wander path and the accuracy of assignment of age of magnetization. It appears that the Orville Coast pole falls on the inferred polar wander path for the East Antarctic craton either considered in situ or rotated to agree with the Lassiter Coast pole. Therefore, an equally plausible explanation for the fact it is distinct from the Lassiter Coast pole is that the mean age of magnetization of the Orville Coast intrusives is rather older than the mean age of magnetization of the Lassiter Coast intrusives and the difference of position is due to apparent polar wander. The density of age determinations on these units is not sufficient to establish the mean age of magnetization for either of these results. Indeed, it seems more probable that such a geographically extensive intrusive complex would record a distribution of directions corresponding with a polar wander path. The resolution of this problem awaits the detailed dating of more of the paleomagnetically sampled igneous units, preferably by the Ar^{40}-Ar^{39} method combined with the observed blocking temperature spectra of the magnetization to demonstrate or discount the hypothesis that the mean directions of magnetization are contemporaneous.

The above discussion leads us to infer that the Cretaceous to Tertiary polar wander path of the Antarctic Peninsula has much the same character as the path

TABLE 3. Poles from Klootwijk and Peirce (1979) Rotated Into Antarctic Reference Frame

Unit (India)	Age m.y.	Pole Position		a_{95}
		Latitude	Longtitude	
Rajmahal traps	102	79°S	173°E	4°
Goro Fm Parh 1st Sanjawi	90	82°S	176°E	3°
Goro Fm Parh 1st Quetta	90	82°S	152°E	4°
Lower Deccan traps	64	72°S	68°E	4°
Upper Deccan traps	61	76°S	68°E	5°
Brewery Island	57	87°S	168°E	22°
Sanjawi Island	55	84°S	204°E	15°
Zam Island	47	82°S	133°E	14°
DSDP core	70	78°S	110°E	..δ
	60	77°S	134°E	11°
	50	85°S	89°E	9°
	40	88°S	182°E	12°
Unit (Australia)				
Bunbury Basalt	100	66°S	197°E	10°
Cygnet complex	98	71°S	200°E	10°
Mt Dromedy (a)	93	82°S	187°E	9°
Mt Dromedy (b)	93	74°S	206°E	10°
Morney Profile	57	84°S	88°E	4°
Older volcanics	55	80°S	191°E	17°
Barrington Volcano	52	86°S	293°E	5°
Liverpool Volcano	34	80°S	21°E	5°

inferred for the East Antarctic craton.

The Cretaceous poles seem to match very well, but the Antarctic early Tertiary poles, in particular, the result from the Snow Island sills, appear displaced from the reference poles of similar age. Any Tertiary tectonic displacement of the Antarctic Peninsula as a whole would have also affected the Cretaceous poles. Hence the positions of the Antarctic Peninsula early Tertiary poles are probably best explained either by unresolved local tectonic displacement, inadequate averaging of secular variation, or departure of the mean early Tertiary paleomagnetic field from an axially central dipole.

SUMMARY AND CONCLUSIONS

The available data from the Antarctic Peninsula allow the tentative construction of a path of apparent polar wander for the mid-Cretaceous through the Tertiary. The problem of the Orville Coast pole in this construction remains to be resolved. However, the excellent agreement of the Orville Coast pole with probably contemporaneous poles from Australia and India rotated into the Antarctic frame of reference would suggest that the Orville Coast rocks were magnetized in their present position.

If the mid-Cretaceous to Tertiary apparent polar wander path of the Antarctic Peninsula is identical to that of the East Antarctic craton, certain conclusions follow:

1. The Antarctic Peninsula down to the Orville Coast has been a part of the Antarctic plate since 100 m.y. ago.
2. Movements of the Antarctic Peninsula required to avoid overlap in reconstructions of Gondwanaland must have occurred before 100 m.y. ago.
3. The problem of the Pacific plate poles discussed by Suarez and Molnar (1980) and Gordon and Cox (1980) remains to be resolved, but any solution must place the Antarctic Peninsula on the Antractic plate.

In regard to the last conclusion, the possibility of a Tertiary plate boundary within West Antarctica cannot be totally discounted because of the lack of paleomagnetic data of sufficient quality and density to construct polar wander paths for other crustal blocks in West Antarctica (Eights Coast, Marie Byrd Land). Paleo-

magnetic data from the Ellsworth Mountain block (Watts and Bramall, 1981) support the idea of post-Cambrian relative movement of this unit with respect to the craton, but the timing of this movement is not constrained.

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